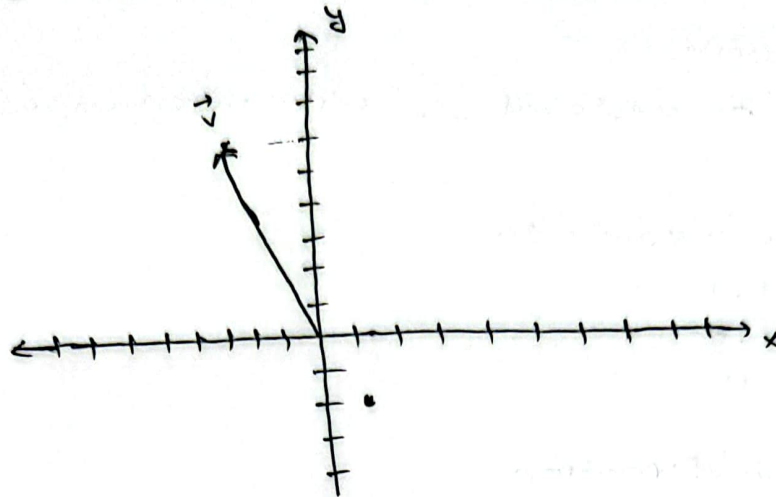


Scalar Multiplication

Q. Vector $\vec{v}$ is graphed in the interactive graph below.

Assuming $-\frac{1}{3}\vec{v}$ starts at the origin, plot its endpoint.



Sol

Multiplying a vector by a scalar is like scaling the directed line segment that represents the vector. This has the following implications:

Magnitude: The magnitude of the scaled vector is equal to the magnitude of the original vector times the absolute value of the scalar. This is represented mathematically as follows:

$$\|c\vec{v}\| = |c| \cdot \|\vec{v}\|$$

Direction: The scaled segment stays on the same line as the original segment. If the factor is positive, the direction stays exactly the same. If the factor is negative, the direction is flipped to the opposite side.

In our case, we are scaling the directed line segment that represents $\vec{v}$ by $-\frac{1}{3}$. So the magnitude of the new segment should be opposite.

To get $\vec{v}$, we have to go 3 units to the left, and 6 units up.

To get $-\frac{1}{3}\vec{v}$, we have to go a third of 3 units to the right (so 1 unit), and a third of 6 units down (so 2 units).

Q. vector $\vec{v}$ has a magnitude of 3 units and it creates an angle of 160° with the positive x-axis.

Find the magnitude and direction of $5\vec{v}$.

Give the direction as an angle measure between 0° and 360° .

$$\|5\vec{v}\| = \text{--- units}$$

$5\vec{v}$ creates an angle of ---° with positive ~~axis~~ ^{x-axis}.

Sol

Finding the magnitude

$$\begin{aligned}\|5\vec{v}\| &= 5\|\vec{v}\| \\ &= 5 \cdot 3 \\ &= 15\end{aligned}$$

Finding the direction

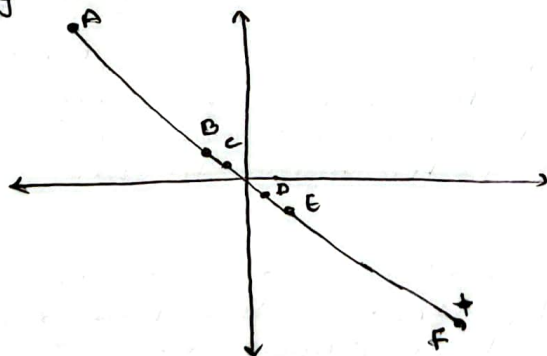
We know $\vec{v}$ creates an angle of 160° with positive x-axis. Since we multiply it by a positive factor, the direction stays the same.

In conclusion, these are the magnitude and direction of $5\vec{v}$:

$$\|5\vec{v}\| = 15 \text{ units}$$

$5\vec{v}$ creates an angle of 160° with positive x-axis.

Q. vector $\vec{v}$ starts at the origin and ends at point E in the drawing below.



Assuming $-5\vec{v}$ starts at the origin, which option could be its endpoint?

Sol

In our case, we are scaling the directed line segment that represents $\vec{v}$ by -5 . So the magnitude of the new segment should be five times that of $\vec{v}$, and the direction should be opposite.

Remember that $\vec{v}$ ends at E. Since $-5\vec{v}$ has five times the magnitude, its possible endpoints are either A or F. Since it is in opposite direction, the only possible option is A.

→ The only option that could be the endpoint of $-5\vec{v}$ is A.